PSD of Line Codes

Wiener Khinchine Theorem

A line code is mathematically represented as.

$$s(t) = \sum_{n=-N}^{N} a_n\, g(t - nT_b)$$

$N \to \infty$

$a_n \to$ statistical, $g(t) \to$ lowpass pulse shaping function.

$$F\{s(t)\} = S(f) = \sum_{n=-N}^{N} a_n\, G(f)\, e^{-j\omega n T_b}$$

$$S(f) = G(f) \sum_{n=-N}^{N} a_n\, e^{-j\omega n T_b}$$

PSD of a random signal $s(t)$ is

$$S_{xx}(f) = \frac{|S(f)|^2}{T}$$

For a random signal $s(t)$ that has $T \to \infty$,

$$S_{xx}(f) = \lim_{T\to\infty} \left[\frac{1}{T} |G(f)|^2\, E \left| \sum_{n=-N}^{N} a_n\, e^{-j\omega n T_b} \right|^2 \right]$$

$$S_{xx}(f) = |G(f)|^2 \lim_{T\to\infty} \left[\frac{1}{T} \sum_{n=-N}^{N} \sum_{m=-N}^{N} E[a_n a_m]\, e^{j(m-n)\omega T_b} \right]$$

ACF $\quad R(k) = E[a_n a_m] = E[a_n a_{n+k}]$

$$= \sum_{i=1}^{L} (a_n a_{n+k})\, P_i$$

Let $T = (2N+1)T_b$, $N \to \infty$, $T \to \infty$

$$S_{xx}(f) = |G(f)|^2 \lim_{N\to\infty} \left[\frac{1}{(2N+1)T_b} \sum_{n=-N}^{N} \sum_{m=-N}^{N} E[a_n a_m]\, e^{j(m-n)\omega T_b} \right]$$

put $m = n+k$ $\quad T = (2N+1)T_b$

$K = m-n$

If $m = -N$ to N, then $k \rightarrow (-N-n)$ to $(N-n)$

$$S_{xx}(f) = |G(f)|^2 \lim_{N \to \infty} \left[\frac{1}{(2N+1)T_b} \sum_{n=-N}^{N} \sum_{K=-N-n}^{N-n} R(K)\, e^{jK\omega T_b} \right]$$

$$= \frac{|G(f)|^2}{T_b} \lim_{N \to \infty} \left[\frac{1}{(2N+1)} \sum_{k=-N-n}^{N-n} R(K)\, e^{jK\omega T_b} \sum_{n=-N}^{N} (1) \right]$$

$$\boxed{S_{xx}(f) = \frac{|G(f)|^2}{T_b} \sum_{K=-\infty}^{\infty} R(K)\, e^{jK\omega T_b}}$$

where $R(K) = \sum_{i=1}^{L} (a_n\, a_{n+K}) \cdot P_i$ $\quad$; $L \rightarrow$ levels

1. Unipolar NRZ

$$s(t) = \begin{cases} A \ ; & 0 \text{ to } T \quad \text{for } 1 \\ 0 \ ; & 0 \text{ to } T \quad \text{for } 0 \end{cases}$$

$$g(t) = \text{Rect}\,(t/T_b)$$

$$G(f) = T_b \,\text{Sinc}\,(\pi f T_b)$$

$$R(k) = \sum_{i=1}^{L} (a_n \cdot a_{n+k})\, P_i$$

$$k=0; \quad R(0) = \sum_{i=1}^{2} a_n \cdot a_n \cdot P_i = 0 \cdot 0 \tfrac{1}{2} + A \cdot A \tfrac{1}{2} = \frac{A^2}{2}$$

$$|k|>0, \quad R(1) = \sum_{i=1}^{L} (a_n \ a_{n+1})\, P_i = \frac{A^2}{4}$$

a_n	a_{n+1}	P_i
0	0	1/4
0	1	1/4
1	0	1/4
1	1	1/4

$$S_{xx}(f) = \frac{|T_b \,\text{Sinc}\,(\pi f T_b)|^2}{T_b} \left[\frac{A^2}{2} + \sum_{\substack{k=-\infty \\ k \neq 0}}^{\infty} A^2/4 \ e^{j2\pi k f T_b} \right]$$

$$= T_b \,\text{Sinc}^2 (\pi f T_b) \left[\frac{A^2}{4} + \frac{A^2}{4} + \sum_{\substack{k=-\infty \\ k \neq 0}}^{\infty} \frac{A^2}{4} e^{jk2\pi f T_b} \right]$$

$$= \frac{A^2}{4} T_b \,\text{Sinc}^2 (\pi f T_b) \left[1 + \sum_{k=-\infty}^{\infty} e^{jk2\pi f T_b} \right]$$

$$\sum_{k=-\infty}^{\infty} e^{jk2\pi f T_b} = \frac{1}{T_b} \sum_{n=-\infty}^{\infty} \delta(f - n/T_b) \quad \text{Train of Impulses.}$$

$$S_{xx}(f) = \frac{A^2 T_b}{4} \,\text{Sinc}^2 (\pi f T_b) \left[1 + \frac{1}{T_b} \sum_{n=-\infty}^{\infty} \delta(f - n/T_b) \right]$$

At $f = n/T_b$, $n \neq 0$,

$$S_{XX}(f) = \frac{A^2 T_b}{4} \operatorname{Sinc}^2(\pi f T_b)\left[1 + \frac{\delta(f)}{T_b}\right]$$

$$= \frac{A^2 T_b}{4}\left[\frac{\sin(\pi f T_b)}{\pi f T_b}\right]^2 \left[1 + \frac{\delta(f)}{T_b}\right]$$

Advantage - Easy to generate, only one power supply is required.

Limitations - 1. Waste of power due to DC level at $f = 0$
2. Clock Extraction is not at peak
3. Long strings of 1's and 0's cause loss of Synchronization
4. No Error detection Capability.

Polar NRZ

$$s(t) = \begin{cases} A & ; 0 \text{ to } T_b \rightarrow 1 \\ -A & ; 0 \text{ to } T_b \rightarrow 0 \end{cases}$$

$$g(t) = Rect(t/T_b)$$

$$R(k) = \sum_{i=1}^{L} (a_n \cdot a_{n+k}) P_i$$

$K=0$, $R(0) = A^2$

a_n	a_n	P_i
A	A	$1/2$
$-A$	$-A$	$1/2$

$= A^2$

$|k| = 1$, $R(1) = 0$

a_n	a_{n+1}	P_i
A	A	$1/4$
$-A$	$-A$	$1/4$
$-A$	A	$1/4$
A	$-A$	$1/4$

$$R(k) = \begin{cases} A^2 & ; K=0 \\ 0 & ; K \neq 0 \end{cases}$$

$$S_{XX}(f) = \frac{|G(f)|^2}{T_b} \sum_{K=-\infty}^{\infty} R(K) e^{j\omega K T_b}$$

$$= A^2 T_b \, Sinc^2(\pi f T_b)$$

Advantage – Reduced DC content and Better Error performance over unipolar

Limitation – ① No error detection capability
② Two power supplies are required
③ Long strings of 0's and 1's results into loss of synchronization.

3. Manchester Signalling

$$S_1(t) = \begin{cases} A & 0 \text{ to } T/2 \\ -A & T/2 \text{ to } T \end{cases} \qquad \text{Symbol 1}$$

$$S_2(t) = \begin{cases} -A & 0 \text{ to } T/2 \\ +A & T/2 \text{ to } T \end{cases} \qquad \text{Symbol 0}$$

$$R(k) = \sum_{i=1}^{L} (a_n \cdot a_{n+k})\, P_i$$

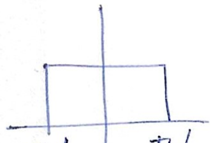
$$k=0, \qquad R(0) = \sum_{i=1}^{2} (a_n \cdot a_n)\, P_i$$

$$= \frac{1}{2}\left[A \cdot A \frac{1}{2} + (-A)(-A)\frac{1}{2} + \frac{1}{2}(-A)(-A) + \frac{1}{2}(A)(A)\right]$$

$$R(0) = \frac{1}{2}\left[\frac{A^2}{2} + \frac{A^2}{2} + \frac{A^2}{2} + \frac{A^2}{2}\right] = A^2$$

$$K \neq 0, \qquad R(0) = 0$$

$$g(t) = \mathrm{Rect}\left(\frac{t + T_b/4}{T_b/2}\right) - \mathrm{Rect}\left(\frac{t - T_b/4}{T_b/2}\right)$$



$$G(f) = \frac{T_b}{2} \operatorname{Sinc}(\pi f T_b/2)\, e^{j\omega T_b/4} - \frac{T_b}{2} \operatorname{Sinc}(\pi f T_b/2)\, e^{-j\omega T_b/4}$$

$$= j T_b \operatorname{Sinc}(\pi f T_b/2) \cdot \sin(2\pi f\, T_b/4)$$

$$S_{xx}(f) = \frac{|G(f)|^2}{T_b} \sum_{k=-\infty}^{\infty} R(k)\, e^{j\omega k T_b}$$

$$= A^2 T_b \operatorname{Sinc}^2(\pi f T_b/2) \sin^2(\pi f T_b/2)$$

Advantages – Zero DC component, No loss of Synchronization

Limitation – No Error detection capability, Large BW is required due to transition.

4. AMI (Alternate Mark Inversion)

$+A$, $-A$ and 0 are used to represent AMI. Its also called Bipolar. Here we use RZ format.

$$g(t) = \text{Rect}\left(t/T_{b/2}\right) = \frac{T_b}{2}\,\text{Sinc}\left(\frac{\pi f T_b}{2}\right)$$

$$R(K) = \sum_{i=1}^{L} (a_n \cdot a_{n+K})\, P_i$$

$$R(0) = \frac{1}{2}(1 \cdot 1) + \frac{1}{2}(0 \cdot 0) = \frac{1}{2}\left[\frac{1}{2}AA + \frac{1}{2}(-A)(-A)\right]$$

$$R(0) = \frac{A^2}{2}$$

$$|K| = 1, \quad R(1) = \sum_{i=1}^{L} (a_n\, a_{n+1})\, P_i$$

a_n	a_{n+1}	P_i	
0	0	$1/4$	$= 0$
0	1	$1/4$	$= 0$
1	0	$1/4$	$= 0$
A	$-A$	$1/4 \times \frac{1}{2}$	$= -A^2/8$
$-A$	A	$\frac{1}{4} \times \frac{1}{2}$	$= -A^2/8$
			$= -A^2/4$

$|K| > 1, \quad R(K) = 0$

$$S_{xx}(f) = \frac{\left|T_{b/2}\,\text{Sinc}\left(\pi f T_{b/2}\right)\right|^2}{T_b}\left[\frac{A^2}{2} - \frac{A^2}{4}e^{j\omega T_b} - \frac{A^2}{4}e^{-j\omega T_b}\right]$$

$$= \frac{A^2 T_b}{8}\,\text{Sinc}^2\left(\frac{\pi f T_b}{2}\right)\left[1 - \frac{1}{2}\left(e^{j\omega T_b} + e^{-j\omega T_b}\right)\right]$$

$$= \frac{A^2 T_b}{8} \text{Sinc}^2\left(\frac{\pi f T_b}{2}\right)\left[1 - \frac{1}{2}\left(e^{j\omega T_b} + e^{-j\omega T_b}\right)\right]$$

$$= \frac{A^2 T_b}{8} \text{Sinc}^2\left(\frac{\pi f T_b}{2}\right)\left[1 - \cos(\omega T_b)\right]$$

$$= \frac{A^2 T_b}{8} \text{Sinc}^2\left(\pi f T_b/2\right)\left[1 - \left(1 - 2\sin^2(\omega T_b/2)\right)\right]$$

$$S_{XX}(f) = \frac{A^2 T_b}{4} \text{Sinc}^2\left(\pi f T_b/2\right) \cdot \sin^2\left(\pi f T_b\right)$$

Advantages - DC Component is zero, Has single error detect Capability.

Limitations - (1) Has loss of Synchronization when long string of zeros transmitted.

(2) 2 power supplies are required

(3) Need 3dB more signal power than a polar signal for same Pe.

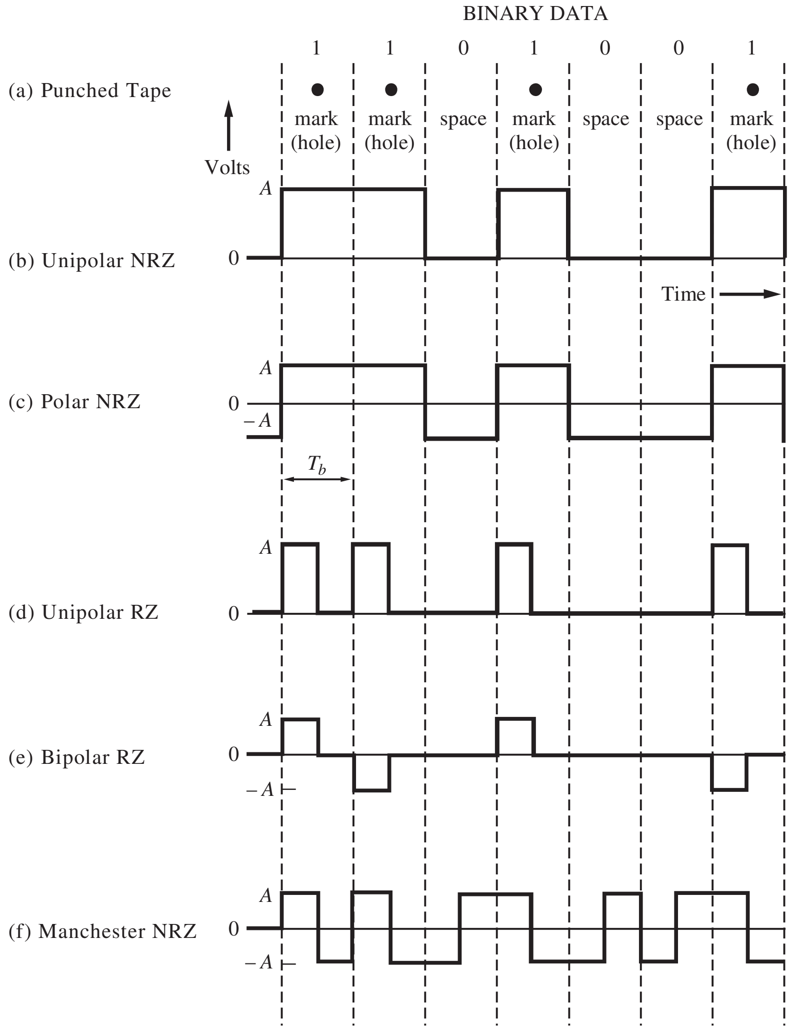


Figure 3–15 Binary signaling formats.

Unipolar Signaling. In positive-logic unipolar signaling, the binary 1 is represented by a high level (+A volts) and a binary 0 by a zero level. This type of signaling is also called *on–off keying*.

Polar Signaling. Binary 1's and 0's are represented by equal positive and negative levels.

Bipolar (Pseudoternary) Signaling. Binary 1's are represented by alternately positive or negative values. The binary 0 is represented by a zero level. The term *pseudoternary* refers to the use of three encoded signal levels to represent two-level (binary) data. This is also called *alternate mark inversion* (AMI) signaling.

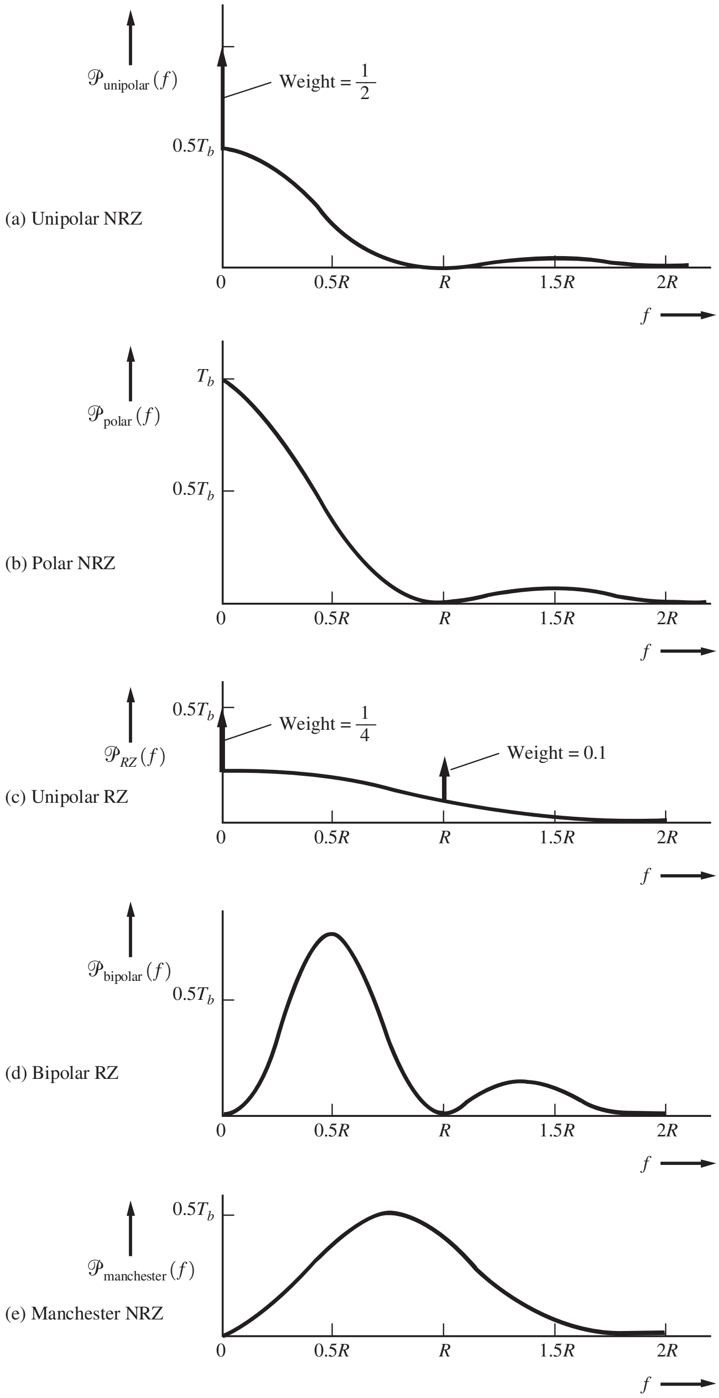


Figure 3–16 PSD for line codes (positive frequencies shown).